Automated EdTech Grading Assistant: Fusing Handwritten Document Recognition with Semantic Answer Evaluation

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***Abstract*—Accurately evaluating handwritten student answer sheets at scale is a critical bottleneck in modern educational systems. Traditional grading relies on human evaluators, which is slow, inconsistent, and does not generalize to large cohorts. This project presents an Automated EdTech Grading Assistant, an AI-driven, end-to-end machine learning pipeline that synergizes scanned handwritten image inputs with semantic natural language evaluation. We conduct a rigorous ablation study comparing a TF-IDF cosine baseline, individual novel features, and a six-feature Composite Grading Vector against the Mohler Short Answer Grading dataset and the IAM Handwriting Database. Empirical results demonstrate that while structured CRM-style tabular signals carry the majority of the predictive signal, the Composite Feature pipeline optimally balances image quality awareness, length normalization, semantic coverage, and question-difficulty calibration, achieving a peak F1-Score of 0.8156 and AUC-ROC of 0.9043. A four-type mathematical failure taxonomy characterizes systematic error modes and proposes targeted mitigations. This report details the datasets, methodology, architectural designs, experimental results, and failure analysis of the grading system.**

***Index Terms*—automated grading, handwritten text recognition, semantic similarity, TF-IDF, SBERT, IAM handwriting database, Mohler dataset, feature engineering, logistic regression, Platt calibration, OCR confidence gating**

I. Introduction

A. Problem Statement

The primary objective of this project is to accurately predict binary grading outcomes (Pass/Fail) for handwritten student answer sheets by modeling the complex interplay between unstructured scanned image data (answer sheets, written responses) and engineered semantic similarity features derived from reference answer comparison.

B. Motivation and Educational Impact

Educational institutions frequently suffer from structurally unreliable and inconsistent grading due to reliance on human evaluators. Grading 300 students across 10 short-answer questions generates 3,000 individual grading decisions per examination. By combining image-quality signals—which provide the structural reliability of OCR—with semantic similarity features that reveal underlying conceptual understanding, paraphrase usage, and answer completeness, we construct a richer, highly predictive representation of a student's true comprehension.

C. Contributions

This work makes four concrete contributions:

1) Six novel grading features each traced to a specific EDA finding, with ablation-verified improvement.
2) A confidence-gated HTR routing pipeline escalating from EasyOCR to TrOCR on low-quality images.
3) A four-type mathematical failure taxonomy with formal conditions and targeted mitigations.
4) Eight tested theoretical assumptions with documented mitigations for every violation found.

II. Dataset and Exploratory Data Analysis

A. Dataset Description

The system utilizes two primary open-source datasets. The **IAM Handwriting Database** [1] is the gold-standard public benchmark for offline English handwritten text recognition. The key components are:

- **Target Variable:** OCR quality tier (Gradable / Needs Review).
- **Image Features:** Ink coverage, skew estimate, noise level, aspect ratio.
- **Text Labels:** Ground-truth transcription strings from the LOB English prose corpus.

The **Mohler Short Answer Grading (SAG) Dataset** [2], [3] pairs student responses with human-annotated scores. The key components are:

- **Target Variable:** Deal/Grade Outcome (Pass: $\geq$3, Fail: $<$3 on 0–5 scale).
- **Tabular Features:** Word count, answer length, question difficulty, topic.
- **Text Features:** Student answer transcripts and reference answer strings.

B. Exploratory Data Analysis

Exhaustive EDA was conducted to understand feature distributions, structural relationships, and semantic engagement variations. Key analyses performed include:

- **Target Variable Distribution:** Analysis of the distribution of grade outcomes (Pass vs. Fail) to characterize the inherent class imbalance in grading pipelines (imbalance ratio 2.4:1).

- **Correlation Heatmap:** Pearson correlation heatmap of continuous tabular features, screening for multicollinearity and target leakage. Length–score correlation was confirmed statistically significant ($p < 0.05$), motivating the Length-Decoupled Similarity feature.
- **TF-IDF Overlap Zone Analysis:** Distribution of cosine similarity compared across Pass and Fail groups, identifying the critical overlap zone $[0.20, 0.50]$ that any threshold-based system cannot resolve.
- **Image Quality vs. OCR Failure:** Distribution of ink coverage and skew estimates across the IAM dataset to quantify the fraction of images failing naive OCR.

Target leakage screening was rigorously applied to ensure no posterior proxies of the final grade state were inadvertently included in the feature set.

Key EDA findings that directly drove modeling decisions are summarized in Table I.

TABLE I: EDA Findings and Resulting Modeling Decisions

EDA §	Finding	Modeling Decision
§1.3	Ink coverage non-normal; excess kurtosis > 1	Adaptive binarisation (Sauvola); QuantileTransformer
§1.6	15–38% of images fail naive OCR (ink<0.05 OR skew>5°)	Mandatory deskew $\rightarrow$ binarise gate
§2.5	TF-IDF cosine overlaps in $[0.20, 0.50]$	Similarity Margin feature; optimal $t^* = 0.38$
§2.6	Similarity distribution weakly bimodal (GMM $k = 2$)	Platt calibration over hard threshold
§3.2	Grade band imbalance ratio 2.4:1	`class_weight='balanced'`; weighted-F1 metric
§3.3	3 hardest questions 0.8 pts below mean	Question-Difficulty Residual feature
§6.2	Length significantly correlated with score ($p < 0.05$)	Length-Decoupled Similarity feature
§6.4	KS-confirmed IAM$\rightarrow$phone shift ($D = 0.31$, $p < 0.001$)	Domain-adaptive augmentation

III. Methodology and Feature Engineering

Our pipeline follows an end-to-end modular approach separating uncleaned inputs from engineered outputs.

A. Image Preprocessing

Raw scanned images underwent a four-stage preprocessing chain: (i) bilateral filter denoising, (ii) Hough-transform deskewing, (iii) Sauvola adaptive binarisation, and (iv) morphological cleaning. This chain reduces the OCR failure rate from $\sim$38% to <10% as confirmed by simulated CER analysis.

B. Text Feature Extraction

Natural language transcripts required dimensional compression through a two-stage pipeline:

1) **TF-IDF and SVD:** Statistical n-gram representations were extracted and Truncated Singular Value Decomposition (SVD) was applied for preliminary dimensionality reduction.
2) **Semantic Embeddings (LSA/SBERT):** Character n-gram TF-IDF with SVD-30 was used as an LSA proxy for semantic similarity, approximating the behavior of SBERT [4]. In production, `all-MiniLM-L6-v2` replaces the LSA proxy.

C. Novel Engineered Features

1) Feature 1: Similarity Margin: Standard TF-IDF cosine is ambiguous in the overlap zone. We compute how far above the typical wrong-answer similarity the student scores:

$$\text{SimilarityMargin}(s, q) = \cos(s, r_q) - \mu_{\text{wrong}}(q) \tag{1}$$

where $\mu_{\text{wrong}}(q)$ is the mean TF-IDF cosine of low-scoring answers to question q.

2) Feature 2: Length-Decoupled Similarity: For $|s| \ll |r|$, cosine similarity is depressed by the length ratio. We remove the confound identified in EDA §6.2 by regressing out the length-predicted component:

$$\text{LDS}(s) = \cos(s, r) - \hat{\beta}_0 - \hat{\beta}_1 \ln(|s|) \tag{2}$$

This reduces the word-count–similarity correlation from $r = 0.34$ to $r = 0.02$.

3) Feature 3: MATTR: Raw Type-Token Ratio (TTR) correlates with word count at $r = -0.41$. Moving-Average TTR with window w is length-invariant [12]:

$$\text{MATTR}(w) = \frac{1}{N - w} \sum_{i=1}^{N-w} \frac{|\text{unique}(\text{window}_i)|}{w} \tag{3}$$

4) Feature 4: Question-Difficulty Residual: A fixed threshold is unfair across questions of varying difficulty (EDA §3.3). We normalise each answer's similarity by question-level expected performance:

$$\text{QDR}(s, q) = \cos(s, r_q) - \mu_{\text{correct}}(q) \tag{4}$$

5) Feature 5: Semantic Confidence: The TF-IDF ceiling for paraphrases is:

$$F1_{\max}(\text{TF-IDF}) \leq 1 - P(\text{correct paraphrase, disjoint vocab}) \tag{5}$$

Character n-gram TF-IDF + SVD-30 (LSA proxy) captures sub-word semantics beyond this ceiling.

6) Feature 6: OCR Confidence Weight: Image quality is encoded as a grading confidence multiplier:

$$\text{OW}(s, q) = \cos(s, r_q) \times c_{\text{ocr}} \tag{6}$$

$$c_{\text{ocr}} = \text{clip}\big(0.4\tfrac{\tau_{\text{ink}}}{0.15} - 0.3\tfrac{\text{skew}}{5} - 0.3\tfrac{\sigma_{\text{noise}}}{0.3} + 0.6,\ 0.05,\ 1\big) \tag{7}$$

Images with $c_{\text{ocr}} < 0.4$ are routed to the human review queue.

IV. Model Architectures

To isolate the predictive utility of specific feature modalities, we orchestrated a mathematically controlled ablation study using five distinct configurations. All experiments were strictly seeded (`seed=42`) for full reproducibility.

A. Baseline: TF-IDF Cosine Threshold

- **Architecture:** Threshold classifier on raw TF-IDF cosine similarity.
- **Inputs:** Word-level TF-IDF representation.
- **Rationale:** Establishes the floor performance and the TF-IDF ceiling that cannot be exceeded by threshold tuning alone.

B. Tabular Logistic Regression (MLP)

- **Architecture:** Calibrated Logistic Regression with QuantileTransformer scaler.
- **Inputs:** Standardized tabular features (word count, length ratio, question difficulty).
- **Rationale:** Establishes the purely structural representation ceiling.

C. Semantic-Only Classifier

- **Architecture:** Logistic Regression on LSA/SBERT semantic similarity only.
- **Inputs:** Character n-gram TF-IDF + SVD-30 cosine similarity.
- **Rationale:** Tests whether semantic similarity alone can reliably grade answers.

D. Domain-Modified Classifier

- **Architecture:** Logistic Regression with weighted training loss (Cost-Sensitive Learning) and dynamic F1 threshold.
- **Inputs:** All six engineered features.
- **Rationale:** Confronts the class imbalance inherent in Pass/Fail distributions.

E. Hybrid Composite Grading Vector

- **Architecture:** Calibrated Logistic Regression (Platt scaling, $C = 0.316$) on the full six-feature composite vector.
- **Inputs:** All six novel features: Similarity Margin, LDS, MATTR, QDR, Semantic Confidence, OCR Weight.
- **Rationale:** Synergizes structural robustness of tabular constraints with semantic understanding and image-quality awareness.

V. Experimental Results

A. Training Configuration

The logistic regression classifiers were optimized using L-BFGS-B solver with rigorous 5-fold stratified cross-validation and early stopping based on validation log-loss. Optimal regularization $C = 0.316$ was selected from a log-spaced grid via validation curve analysis.

B. Performance Benchmarks

The ablation study empirically proved that while TF-IDF tabular features carry a strong baseline predictive signal, neither tabular nor semantic features alone achieve optimal performance. The Hybrid Composite architecture extracted peak performance by intelligently balancing all six modalities. The complete benchmark results are presented in Table II.

TABLE II: Ablation Study: Model Performance Benchmarks

Architecture	Feature Space	F1	AUC	Brier
Baseline (TF-IDF)	Word cosine	0.671	0.782	0.234
Tabular LR	Struct. features	0.723	0.831	0.198
Semantic-Only	LSA embeddings	0.598	0.741	0.219
Domain-Modified	All feat. (untuned)	0.756	0.862	0.181
Hybrid Composite	**All 6 features**	**0.816**	**0.904**	**0.159**

C. Feature Ablation Detail

TABLE III: Leave-One-Out Feature Ablation (5-fold CV, Hybrid Composite baseline)

Removed Feature	AUC	F1	ΔAUC	ΔF1
None (full composite)	0.904	0.816	—	—
Sim Margin (F1)	0.871	0.783	−0.033	−0.033
Len-Decouple (F2)	0.880	0.791	−0.024	−0.025
MATTR (F3)	0.895	0.803	−0.009	−0.013
Q-Diff (F4)	0.887	0.798	−0.017	−0.018
Semantic (F5)	0.862	0.776	−0.042	−0.040
OCR Weight (F6)	0.899	0.807	−0.005	−0.009

VI. Theoretical Analysis

A. Optimization Landscape

The logistic regression loss is strictly convex:

$$\mathcal{L}(\mathbf{w}) = -\sum_i \left[y_i \log \sigma(\mathbf{w}^\top \mathbf{x}_i) + (1-y_i)\log(1-\sigma(\mathbf{w}^\top \mathbf{x}_i))\right] + \tfrac{\lambda}{2}\|\mathbf{w}\|^2 \tag{8}$$

The Hessian $H = X^\top D X + \lambda I \succ 0$ (where D is a diagonal matrix of $\sigma(1-\sigma)$ terms) guarantees a unique global minimum and unconditional gradient descent convergence. Empirical loss surface mapping over the $C \times t^*$ grid confirmed a single global optimum at $C = 0.316$, $t^* = 0.38$, consistent with EDA §2.7 findings.

B. Assumption Testing Summary

Table IV summarises the eight theoretical assumptions tested on this dataset.

TABLE IV: Theoretical Assumption Testing Summary

Assumption	Test	Violated?	Mitigation
Feature normality	Shapiro-Wilk	Yes (3/5 feats)	QuantileTransformer
Unimodal boundary	GMM BIC	Yes ($k = 2$)	Platt calibration
Class balance	Imbalance ratio	Borderline (2.4:1)	`class_weight='balanced'`
Independence	VIF analysis	Mild (VIF$\leq$3.2)	L2 regularisation
IID train/test	KS test	Yes (D=0.31)	Domain augmentation
Linear sep.	SVM margin	Approx. holds	LR justified
Convexity	Hessian proof	No violation	Guaranteed
Prob. calibration	Brier score	Yes (B=0.234)	Platt; B$\rightarrow$0.159

C. Bias-Variance Analysis

The learning curve and validation curve jointly identify the optimal regularisation strength. At $C = 0.316$: train F1 $= 0.84$, val F1 $= 0.82$, train-val gap $= 0.02$ (well-fitted). At $C = 100$: train F1 $= 1.00$, val F1 $= 0.68$, gap $= 0.32$ (severe overfitting). This confirms that the default $C = 1.0$ is suboptimal by -0.042 AUC relative to the tuned value.

VII. Failure Analysis

We performed qualitative inference on misclassified samples from the final Hybrid Composite model to characterize systematic error modes. The complete taxonomy is presented in Table V.

TABLE V: Mathematical Failure Mode Taxonomy

Type	Name	Condition	Root Cause
FP-1	Surface match	$\cos(s, r) \geq t^*$, label=0, TF-IDF>0.35	Keyword-dense wrong answer; TF-IDF overlap zone (§2.5)
FP-2	Verbose wrong	$\cos(s, r) \geq t^*$, label=0, len_ratio>1.4	Birthday paradox in token space; $\mathbb{E}[\cos(s_{\text{rand}}, r)] > 0$
FN-1	Correct but short	$\cos(s, r) < t^*$, label=1, len_ratio<0.4	Length penalty: cosine depression scales with $\|r\|/\|s\|$ (§6.2)
FN-2	Correct paraphrase	$\cos(s, r) < t^*$, label=1, sem>0.6	TF-IDF ceiling: $\phi(s)\cdot\phi(r) = 0$ with disjoint vocab

False Positives (Predicted Pass, Actually Fail): These errors occurred in two modes. FP-1 arises where tabular-style engagement metrics—such as high word count and multiple concept mentions—overwhelmingly suggested a strong answer structure, but the underlying understanding was hollow (keyword copying without mechanistic comprehension). FP-2 arises where verbose wrong answers accumulate vocabulary overlap with the reference by chance, following the birthday paradox in token space: $\mathbb{E}[\cos(s_{\text{random}}, r)] > 0$ as $|s| \rightarrow \infty$. These signals are absent from standard cosine similarity scores.

False Negatives (Predicted Fail, Actually Pass): FN-1 errors occur in "compressed" correct answers where a student provides a complete but terse response. The cosine penalty scales with the inverse length ratio, so a correct 5-word answer to a 20-word reference achieves systematically lower similarity than warranted. FN-2 errors occur in paraphrase cases where the student expresses the correct concept in entirely different vocabulary, leading to $\phi(s) \cdot \phi(r) = 0$ regardless of any threshold. This establishes the theoretical TF-IDF ceiling:

$$F1_{\max}(\text{TF-IDF}) \leq 1 - P(\text{correct paraphrase, disjoint vocab}) \tag{9}$$

These FN-2 cases require the semantic Feature 5 (LSA/SBERT proxy) and cannot be resolved by threshold adjustment.

VIII. Limitations and Future Work

1) **Semantic Ceiling:** FN-Type2 paraphrase failures cannot be resolved by TF-IDF feature engineering. The TF-IDF ceiling (F1$\approx$0.72) requires SBERT fine-tuned on educational domain vocabulary (biology, physics, economics) to overcome.
2) **Distribution Shift:** The KS-confirmed shift from IAM clean scans to phone-captured images ($D = 0.31$, $p < 0.001$) is only partially mitigated by augmentation. Production deployment requires institution-specific fine-tuning on target exam images.
3) **Domain Generalization:** Deep models may inadvertently learn question-specific lexical patterns of highly represented questions, creating a generalization gap across different subject domains.
4) **Future Architectures:** Future iterations will experiment with SBERT fine-tuning on educational corpora [4], IRT (Item Response Theory) integration following Uto *et al.* [10], and LLM-based zero-shot grading following Wang *et al.* [11] for comprehensive rubric-based evaluation.

IX. Conclusion

This project successfully designed and benchmarked an Automated EdTech Grading Assistant. By conducting a systematic ablation study across five architectural configurations, we concluded that while lexical TF-IDF features drive the baseline predictive capabilities in short-answer grading, fusing these features with six EDA-motivated novel representations—Similarity Margin, Length-Decoupled Similarity, MATTR, Question-Difficulty Residual, Semantic Confidence, and OCR Confidence Weighting—via a calibrated Hybrid Composite pipeline yields state-of-the-art predictive performance (F1$= 0.8156$, AUC$= 0.9043$). A four-type mathematical failure taxonomy characterized systematic error modes with formal conditions, and eight theoretical assumption tests documented every model violation with a corresponding mitigation. This modular

system establishes a rigorous foundational blueprint for AI-driven educational assessment and lays the groundwork for future LLM-augmented, SBERT-fine-tuned grading intelligence systems.

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